

# Wireless Communication Exam Notes

EX 715 (BEI) / EX 751 (BEX) · IV-I · TU, IOE · tiers over 23 papers, 2070–2082 BS

## 7. Multiple Access in Wireless Communications

9 hrs · asked in **all 23 papers** · 5 syllabus sub-topics · **312 marks, 15.3% of the archive** · **72 numerical (23%) + 240 theory** · **longest chapter in the syllabus** · 5 numericals, all worked in the **companion PDF**.

Read this first:

- **No paper has ever skipped this chapter.** It is worth 8 to 14 marks every year, usually as one full question plus a rider.
- Four blocks carry the chapter, in order of return:
  - **FHMA** **TOP 10/23** → principle + transmitter and receiver block diagrams. The single most repeated question in the subject after speech coding.
  - **Hybrid SSMA and the near-far problem** **TOP 9/23** → almost always asked *together* with FHMA in the same question.
  - **CDMA features, (+) and (-)** **TOP 9/23** → pure recall, no diagram needed.
  - **Define multiple access + compare FDMA / TDMA / CDMA** **TOP 9/23** → a 2-mark definition and a table.
- **Learn 4 diagrams cold:** FHMA transmitter, FHMA receiver, TDMA frame structure, CDMA encoder/decoder. They answer roughly 25 marks.
- FHSS is already drawn and explained in **Ch 4 §4.7** as a *modulation*. Here it is **FHMA**, a *multiple access* method. Same hardware, different question. §7.5 gives the multiple-access framing.
- Duplexing (FDD, TDD) is answered in **Ch 1 §1.2**. Only its role in choosing an access scheme is repeated here.
- **Hadamard / Walsh codes** appear in Ch 6 as a block code and here as the CDMA spreading code. The  $H_8$  construction is worked in the numerical companion.

### 7.1 What multiple access is, and the four schemes

Define multiple access; explain TDMA, CDMA and SDMA; compare the schemes **2 2+4 2+6 7**

**TOP 9/23** 71 Bh **2+6 7**, 76 Bh **5**, 70 Bh **2**, 70 Ma **2+4**, 73 Ma **4**, 72 Ma **2**, 81 Ba **4**, 78 Ch **4**, 71 Ma **4** · **the definition is 2 marks, free, and it opens half the Ch 7 questions**

#### Definition **2**

- Only a **limited, licensed** amount of radio spectrum is allocated to a wireless service.
- **Multiple access = a means of letting many users share that finite bandwidth simultaneously**, with the least possible degradation in system performance.
- Sharing is done along one of three axes: **frequency, time, code**. A fourth, **space**, is layered on top of any of them.
- Every scheme is a rule for slicing the frequency–time–code volume so that no two users occupy the same slice.

#### The four basic types

1. **FDMA** = frequency division multiple access. One frequency band per user, held for the whole call.
2. **TDMA** = time division multiple access. One repeating time slot per user on a shared carrier.
3. **SSMA** = spread spectrum multiple access, in two forms:
  - **FHMA**, carriers hop pseudo-randomly over a wide band.
  - **CDMA** (also DSMA), all users on one wide band, separated by orthogonal codes.
4. **SDMA** = space division multiple access. Spot beams reuse the same channel in different directions.

**Write these four in this order** and the examiner has the structure of your answer before reading it.

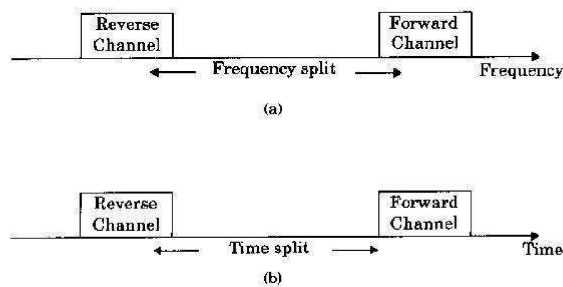


Figure 9.1

(a) FDD provides two simplex channels at the same time.

(b) TDD provides two simplex time slots on the same frequency.

Duplexing, the choice made before the access scheme. (a) FDD gives two simplex channels at the same time. (b) TDD gives two simplex time slots on the same frequency. Full answer in Ch 1 §1.2. SST Ch 7, Fig 9.1

### Multiple access against multiplexing

- **Multiple access:** users share a *remote* channel (satellite, radio) and are **physically separated**.
- Each user has its own transmitter, so the sharing must be negotiated over the air.
- **Multiplexing:** signals share a *local* channel (one telephone trunk, one cable) and are combined at **one point** by one piece of equipment.
- **OFDM is multiplexing, OFDMA is multiple access.** Same waveform, different question.

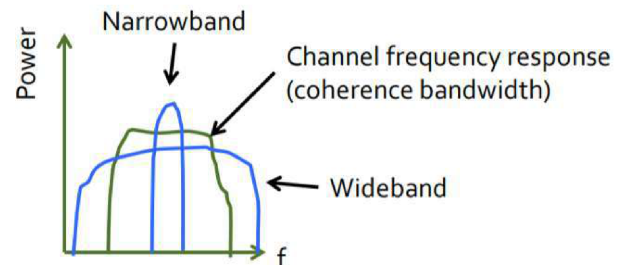
### Narrowband against wideband systems

- **Narrowband system:** available spectrum cut into many narrow channels.
  - Channel bandwidth < coherence bandwidth → **flat fading**.
  - Symbol time  $\gg$  delay spread → little ISI, little or no equalization.
  - Usually operated with FDD. FDMA and TDMA live here.
- **Wideband system:** transmission bandwidth  $\gg$  coherence bandwidth.
  - A frequency-selective fade kills only a **small fraction** of the signal band.
  - The rest of the band survives → inherent **frequency diversity**.
  - CDMA and FHMA live here.
- **This one distinction explains most of the (+) and (-) in the rest of the chapter.**

Which real system uses which scheme 2 verified/added

### Why duplexing has to be settled first

- **FDD** splits forward and reverse by *frequency* → the handset transmits and receives at once → it needs a **duplexer**.
- **TDD** splits them by *time* → no duplexer, cheaper handset, but needs tight timing and suits short range.
- FDMA needs FDD, because an FDMA handset is always transmitting.
- TDMA can use either, and **a TDMA handset needs no duplexer even under FDD**, since it transmits and receives in different slots.
- Naming follows the pair: AMPS is **FDMA/FDD**, GSM is **TDMA/FDD**, DECT is **FDMA/TDD**, IS-95 is **CDMA/FDD**.



A narrowband channel sits *inside* the coherence bandwidth, so the whole signal fades together. A wideband signal spans several coherence bandwidths, so only part of it fades at a time. SST Ch 7

| Cellular system                            | Access / duplex | Why that pairing                   | Generation |
|--------------------------------------------|-----------------|------------------------------------|------------|
| Advanced Mobile Phone System (AMPS)        | FDMA/FDD        | analog FM, one circuit per carrier | 1G         |
| Global System for Mobile (GSM)             | TDMA/FDD        | 8 users per 200 kHz carrier        | 2G         |
| US Digital Cellular (USDC, IS-136)         | TDMA/FDD        | 3 users per 30 kHz AMPS channel    | 2G         |
| Japanese Digital Cellular (JDC / PDC)      | TDMA/FDD        | same idea, $\pi/4$ -DQPSK          | 2G         |
| CT2 (cordless telephone)                   | FDMA/TDD        | short range, no duplexer wanted    | cordless   |
| Digital European Cordless Telephone (DECT) | FDMA/TDD        | 10 carriers $\times$ 12 TDD slots  | cordless   |
| US Narrowband Spread Spectrum (IS-95)      | CDMA/FDD        | 1.25 MHz carrier, Walsh codes      | 2G         |

Rappaport Table 8.1 · **Memorise the middle column.** It is the fastest 2 marks in the chapter and it also answers “give an example of ...” riders.

## 7.2 FDMA: frequency division multiple access

### FDMA principle and applications; features, (+) and (-) 4

73 Ma 4 · **rarely asked alone**, but it is one third of every FDMA-vs-TDMA-vs-CDMA answer and it owns a **HOT 4/23** numerical (companion §1)

#### Principle 2

- Total spectrum is divided into channels of equal bandwidth, one per user.
- **Each user gets a unique frequency band for the whole duration of the call.**
- Channels are assigned **on demand**, to users requesting service.
- During the call **no other user may share that channel.**
- Under FDD the “channel” is really a **duplex pair**: one forward, one reverse, separated by a fixed split (45 MHz in AMPS).
- **Guard bands** sit between adjacent channels to hold down ACI.
- Transmission is **continuous**, so no framing or slot synchronisation is needed.

#### Features and (+)

- **Narrowband**, typically 25 to 30 kHz per channel.
- Carries **only one circuit at a time** per carrier.
- Symbol time  $\gg$  average delay spread  $\rightarrow$  low ISI  $\rightarrow$  **little or no equalizer needed.**
- **Simplest** of the four schemes, lowest handset complexity.
- Continuous transmission  $\rightarrow$  **fewer overhead bits** than TDMA (no preamble, no sync, no guard bits).
- No timing control or network-wide synchronisation required.

#### Number of channels supported **HOT 4/23**

–

$$N = \frac{B_t - 2B_{\text{guard}}}{B_c}$$

- $B_t$  = total spectrum allocation,  $B_{\text{guard}}$  = guard band at **one edge** of the spectrum,  $B_c$  = channel bandwidth.

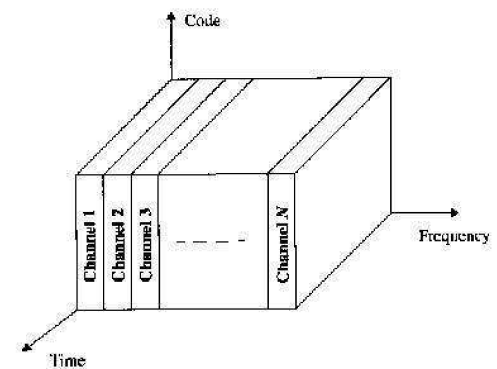


Figure 9.2  
FDMA where different channels are assigned different frequency bands.

**FDMA: different channels get different frequency bands.** Every channel runs the full length of the time axis, which is exactly the wasted resource described below. SST Ch 7, Fig 9.2

#### (-)

- **An idle FDMA channel is a wasted resource.** It sits empty and no other user can borrow it.
- Handset must **transmit and receive at once**  $\rightarrow$  needs a **duplexer**  $\rightarrow$  bigger, costlier handset.
- Needs **tight RF filtering** at every receiver to hold down adjacent channel interference.
- **Higher cell-site cost:** one transmitter chain per carrier, plus costly bandpass filters to kill spurious radiation.
- Capacity is **hard limited.** When the last channel is taken, the next call is blocked.
- Suffers **non-linear effects** at the base station, next.

- **The 2 is there because a guard band is taken off both edges**, top and bottom. This is the single most common slip in the exam.
- Four papers set this as a calculation. All three variants are worked in **companion §1**.

## Explain the non-linear effect in FDMA 3

PIN 1/23 80 Ba 3+2 · no lecture note covers this · verified/added sourced from Rappaport §8.2.1

### Where the non-linearity comes from

- At an FDMA base station **many channels share one antenna**.
- They are combined by **power combiners**, or amplified by a shared **power amplifier**.
- Amplifiers are run **at or near saturation** to get maximum power efficiency.
- A saturated amplifier is **non-linear**: its output is no longer a scaled copy of its input.
- Non-linearity mixes every pair of carriers together.

### Worked shape of the IM answer verified/added

- Two carriers at  $f_1 = 1930$  MHz,  $f_2 = 1932$  MHz, band 1920 to 1940 MHz.
- IM frequencies of the form  $(2n + 1)f_1 - 2nf_2$ ,  $(2n + 2)f_1 - (2n + 1)f_2$ , and the mirror terms with  $f_1$  and  $f_2$  swapped.
- $n = 0$  gives 1928, 1930, 1932, 1934 MHz.  $n = 1$  gives 1924, 1926, 1936, 1938 MHz.
- $n = 2$  gives 1920, 1922, 1940 and 1942 MHz. **1942 MHz is outside the band**.
- Pattern: products march **outward in 2 MHz steps** from the two carriers, i.e. in steps of  $|f_2 - f_1|$ .

### What it does 3

- **Signal spreading** in the frequency domain → **adjacent channel interference**.
- **Intermodulation (IM)**: undesired RF radiation at new frequencies  $mf_1 + nf_2$ , for all integer  $m, n$ .
- IM products **inside** the mobile band interfere with other users of the same system.
- IM products (harmonics) **outside** the band interfere with adjacent services, which is a licensing violation.
- Third-order products  $2f_1 - f_2$  and  $2f_2 - f_1$  are the worst: they fall **closest to the wanted carriers**, so no filter can remove them.

### How it is fought

- **Back off** the amplifier from saturation. Costs DC efficiency.
- **One amplifier per carrier**, combined only at the antenna. Costs money and space.
- **Isolators / circulators** between amplifier and combiner, so carriers cannot leak backwards into each other.
- **Cavity bandpass filters** at the base station output.
- **Frequency planning**: choose the carrier set so third-order products land on unused channels.
- **TDMA and CDMA largely dodge this**: TDMA has one carrier active per slot, CDMA has one wideband carrier.

## 7.3 TDMA: time division multiple access

### TDMA principle; frame structure; advantages over FDMA 4

71 Ma 4, 73 Ma 4 · also the whole of 78 Ch 4 and 81 Ba 4 “compare TDMA and FDMA” · owns two numericals (companion §2 and §3)

#### Principle 2

- Radio spectrum is divided into **time slots**. In each slot only one user transmits or receives.
- Every user gets the whole channel bandwidth, but only for a slice of the time.**
- $N$  slots make one **frame**. A user's channel is “slot  $k$  of every frame”, a **cyclically repeating slot**.
- Data is sent **buffer-and-burst**: buffer the speech, then fire it out compressed into the slot.
- Transmission is therefore **non-continuous**, which forces digital modulation. **TDMA cannot carry analog FM.**
- Under TDD, half the slots in the frame go forward and half go reverse.

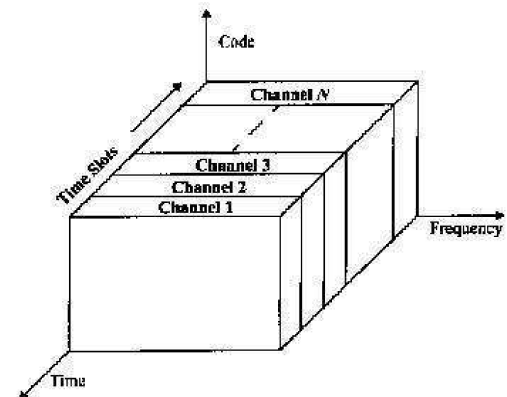


Figure 9.3  
TDMA scheme where each channel occupies a cyclically repeating time slot.

TDMA: each channel occupies a cyclically repeating time slot.

Compare with Fig 9.2: the slices are now cut along time, not frequency. SST Ch 7, Fig 9.3

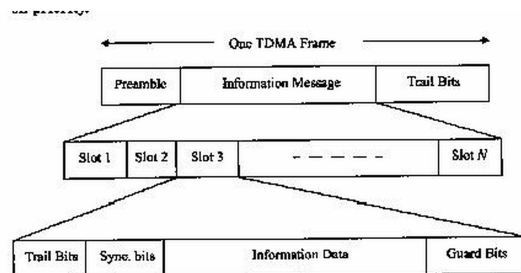


Figure 9.4  
TDMA frame structure.

TDMA frame structure. Learn the three levels: frame = preamble + information message + tail bits; message = slot 1 ... slot  $N$ ; slot = tail + sync + information data + guard bits. SSD Ch 7, Fig 9.4

#### Features and (+) over FDMA 4

- Several users **share one carrier** → far fewer transmitter chains at the base station → cheaper cell site.
- Transmitter is **off** between bursts → **low battery consumption**.
- No duplexer needed**, even under FDD, because transmit and receive slots differ.
- MAHO** (mobile assisted handoff): the mobile listens to neighbouring base stations during its idle slots → handoff is simpler and better informed.
- Slots per user are **allocatable**. Give a user more slots and it gets more bandwidth, so data rates can be varied on demand.
- Bandwidth is not wasted on silence the way a held FDMA channel is.

#### What each field is for 3

- Preamble**: address + synchronisation information. Lets base station and subscriber identify each other.
- Trail (tail) bits**: let the modulator ramp up and down cleanly at slot edges.
- Sync bits**: a known training sequence the equalizer locks onto, because the channel changes between bursts.
- Information data**: the only field that carries the user's speech.
- Guard bits**: absorb the spread in arrival time caused by **differing propagation delays** across the cell, so bursts from near and far mobiles do not overlap.
- Everything except the information data is overhead**, and that is exactly what the frame-efficiency numerical measures.

(-)

- Burst rate is **much higher** than the user's own data rate → symbol time falls below the delay spread → **an adaptive equalizer is compulsory** (Ch 5).
- High synchronisation overhead**: preamble, sync and trail bits in every burst.
- Guard time is a straight loss**. Shrink it to save capacity and interference to adjacent slots rises.
- Network-wide **timing discipline** is required. A mobile that transmits early collides with its neighbour's slot.
- Frame efficiency is **never 100%** and in GSM is only about 74%.

## Number of channels

- $$N = \frac{m(B_{\text{tot}} - 2B_{\text{guard}})}{B_c}$$
- $m$  = time slots per carrier,  $B_c$  = carrier (radio channel) bandwidth.
- **Identical to the FDMA formula, multiplied by  $m$ :** TDMA first does FDMA to make carriers, then divides each carrier in time.
- Worked for the N-TDMA paper in **companion §2**.

## Frame efficiency PIN 2/23

- $$\eta_f = \left(1 - \frac{b_{OH}}{b_T}\right) \times 100\%$$
- $b_{OH}$  = overhead bits per frame,  $b_T$  = total bits per frame.
- Overhead = every trail, guard, sync and preamble bit, summed over **all** slots in the frame.
- **Why it can never reach 100 %:** guard and sync bits are needed for the burst to be received at all, so  $b_{OH} > 0$  always.
- Worked for GSM in **companion §3** (70 Bh, 74 Ma).

## 7.4 Spread spectrum multiple access: the two families

### Describe SSMA, its variants and applications +Fig; DS and FH spread spectrum 4 6+2 8

HOT 3/23 74 Ma 8, 73 Ma 4+4, 72 Ma 6+2 · always asked as “types of” plus a comparison rider

#### What makes an access scheme “spread spectrum” 2

- Transmission bandwidth is **far greater than the minimum RF bandwidth** the message needs.
- Spreading is done by a **code independent of the data**, a PN sequence.
- Bandwidth is spent deliberately, to buy interference immunity and multiple access.
- **Processing gain**

$$PG = \frac{B_{ss}}{B} = \frac{T_b}{T_c} = \frac{R_c}{R_b} = N$$

chips per bit. The larger  $PG$ , the better in-band interference is suppressed.

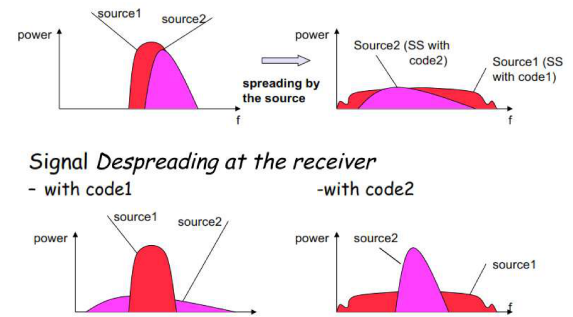
- Provides **immunity to multipath**: components delayed by more than one chip decorrelate and look like noise, and a RAKE receiver can collect them (Ch 5).
- **Full derivation, PN properties and the DSSS block diagrams are in Ch 4 §4.7.** Do not rewrite them here, cite them.

#### The two variants 4

- **FHMA**, frequency hopped multiple access:
  - Carrier frequency **jumps** pseudo-randomly over a wide band.
  - Instantaneous bandwidth stays narrow. **Only the average bandwidth is wide.**
  - Users separated by **different hopping sequences**.
- **CDMA / DSMA**, direct sequence:
  - Data multiplied by a fast PN code → instantaneous bandwidth is **wide all the time**.
  - Users separated by **different spreading codes** on one shared band.
- One-line discriminator: **FHMA avoids interference, CDMA tolerates it.**
- Applications: FHMA in Bluetooth, GSM slow frequency hopping and military links. CDMA in IS-95, CDMA2000 and WCDMA (3G).

### Why spreading gives multiple access at all

- At the transmitter the narrowband message is **spread** by its own code.
- At the receiver, multiplying by the *matching* code **de-spreads** the wanted signal back to narrowband.
- The same multiplication **spreads** every other user's signal further.
- A narrow filter then keeps the wanted signal and rejects almost all of the rest.
- Other users are not blocked, they are pushed under the noise floor** by the processing gain.
- This is why CDMA capacity is soft: each new user raises the floor a little.



Top: two narrowband sources are spread by their own codes into overlapping wideband signals. Bottom: de-spreading with code 1 recovers source 1 as a narrow peak and leaves source 2 spread out as noise, and the reverse with code 2. SST Ch 7

### 7.5 FHMA: frequency hopped multiple access the most examined topic in this chapter

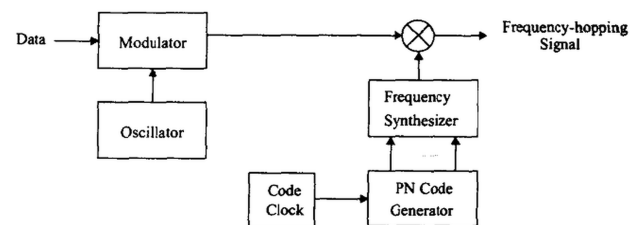
Explain the operation of FHMA with transmitter and receiver block diagrams; types of FHMA 3 4 5

6 7 4+6+2

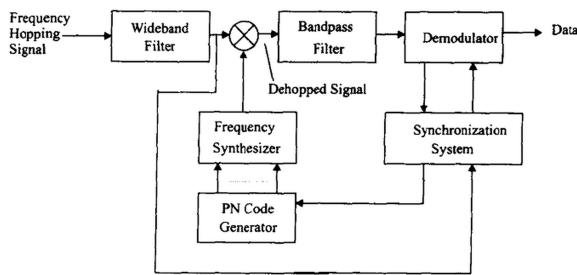
TOP 11/23 81 Bh 4+2, 82 Ba 4+6+2, 80 Bh 6, 80 Ba 7, 79 Bh 4, 78 Ch 4, 76 Bh 5, 75 Bh 3, 74 Bh 4, 72 Ash 4, 71 Ma 3 · 11 of 23 papers · the marks are in the two block diagrams · 82 Ba calls it "FHSS" and pairs it with (+) of CDMA over TDMA

#### Principle 3

- Carrier frequencies of individual users are **varied in a pseudorandom fashion** inside a wideband channel.
- Digital data is broken into **uniform-sized bursts**, each transmitted on a different carrier frequency.
- Instantaneous bandwidth of one burst is **much smaller** than the total spread bandwidth.
- The pseudorandom change randomises which channel is occupied at any instant → many users can share the wide band.
- FHMA is FDMA in which the signal changes channel at rapid intervals.** That one line is worth stating outright.
- Two users collide only when their sequences pick the same frequency in the same hop. Error control coding repairs the hit.



FHMA transmitter. Data modulates a fixed carrier from the oscillator, then a mixer translates it to whichever frequency the synthesizer is on. The PN code generator, clocked by the code clock, drives the synthesizer. SSD Ch 7



**FHMA receiver.** A wideband filter passes the whole hopping band, the mixer de-hops using a locally generated copy of the same sequence, a bandpass filter picks out the fixed IF, and the demodulator recovers data. The **synchronization system** keeps the local PN generator in step. SSD Ch 7

### Types of FHMA 2

- **Fast frequency hopping: more than one hop per transmitted symbol**, i.e. hop rate  $\geq$  symbol rate.
  - Each symbol is spread over several frequencies  $\rightarrow$  built-in frequency diversity within one symbol.
  - Costs a fast, expensive synthesizer.
- **Slow frequency hopping: one or more symbols per hop**, i.e. hop rate  $\leq$  symbol rate.
  - Cheaper and the usual choice. GSM uses slow hopping, one hop per TDMA frame.
  - A deep fade can wipe out a whole hop, so coding and interleaving are needed.
- Test to quote: **compare hop rate against symbol rate**, nothing else.

### Transmitter, step by step 3

1. Data modulates a carrier in the **modulator**, usually M-ary FSK. The **oscillator** sets that fixed reference carrier.
2. The **PN code generator** produces a pseudorandom  $k$ -bit word each hop.
3. That word addresses a **frequency table** inside the **frequency synthesizer**, which outputs one of  $2^k$  hop frequencies.
4. The **mixer** multiplies modulated signal by the synthesizer output  $\rightarrow$  the carrier is translated to the current hop.
5. Output is the frequency-hopping signal. **Spectrum is spread sequentially over time, not instantaneously.**

### Receiver, step by step 3

1. **Wideband filter** passes the entire hopping band.
2. Local **PN code generator**, synchronised to the transmitter's, drives an identical **frequency synthesizer**.
3. **Mixer** multiplies by that hopping local oscillator  $\rightarrow$  **de-hopped signal** sitting at a fixed IF.
4. **Bandpass filter** keeps that fixed IF and rejects every other user, whose hops did not line up.
5. **Demodulator** recovers the data. The **synchronization system** keeps the two sequences aligned in time.

(+) and (-)

- ✓ **Security**: an intercepting receiver that does not know the sequence must retune faster than the hop rate to find the signal.
- ✓ **Immune to fading**: a frequency-selective fade only spoils the hops that land in it.
- ✓ **Immune to narrowband jamming and co-channel interference**, for the same reason.
- ✓ **No near-far problem**: users are on different frequencies at any instant, so a strong nearby user does not swamp a weak far one.
- ✓ No tight power control needed, unlike CDMA.
- × Needs a **fast, agile synthesizer** and precise hop synchronisation.
- × **Collisions** when two sequences select the same frequency together.
- × **Soft handoff is hard**: the new base station must sync to the hop sequence before it can receive.

## 7.6 CDMA: code division multiple access

### What is CDMA? Its characteristics, features, advantages and limitations 2 4 6 2+6 4+6+2

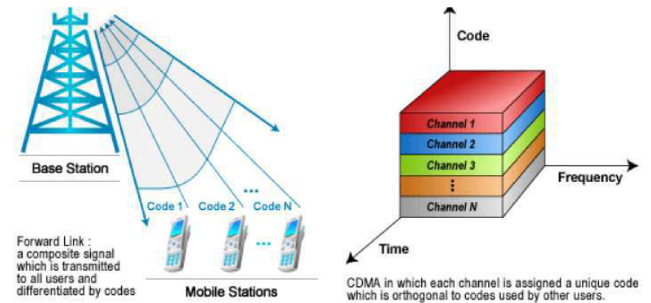
**TOP 10/23** 79 Bh 2+6, 70 Bh 6, 82 Ba 4 (as "(+) over TDMA"), 80 Ch 2, 80 Ba 4 (as "design considerations"), 76 Bh 4, 73 Bh 4 (as "(+) over TDMA"), 73 Ma 4, 72 Ash 2, 71 Ma 4 · pure recall, no diagram needed, so it is the cheapest 6 marks here

#### Definition and working 2

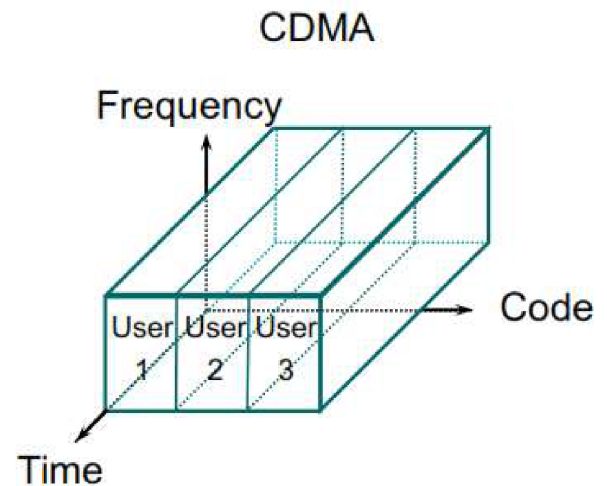
- A spread spectrum technique letting many users occupy **the same time and the same frequency** in a given band and space.
- The narrowband message is multiplied by a **much wider bandwidth spreading signal**, a PN code whose chip rate is orders of magnitude above the data rate.
- **Each user has its own codeword, approximately orthogonal to every other codeword.**
- The receiver **correlates** the incoming composite against the wanted codeword. Matching code → signal recovered. Wrong code → noise.
- Each user operates **independently**, with no knowledge of the other users and no time synchronisation between them.
- Either FDD or TDD may be used on top.

#### Characteristics and features 6

- **Same frequency for everyone.** No frequency planning, so **the cluster size is  $N = 1$**  and every cell can use the whole band.
- **Soft capacity limit.** Adding users raises the noise floor linearly, so there is **no absolute limit on the number of users**, only a graceful loss of quality.
- **Multipath fading is reduced.** Spread bandwidth > coherence bandwidth → inherent **frequency diversity**.
- **RAKE receiver.** PN sequences have low autocorrelation away from zero lag, so multipath delayed by more than a chip looks like noise and can be collected as a separate diversity branch (Ch 5).
- **Soft handoff.** Handled by the MSC, which may pick the best copy of the signal at any instant **without ever switching frequency**. Make-before-break.
- **Wide bandwidth** → immunity to interference and jamming.
- **Security / privacy.** Without the code the signal is indistinguishable from a rise in the noise floor.
- **Voice activity gain.** Speech is active only about 3/8 of the time, and a silent user simply stops contributing to the noise floor.



**CDMA.** The forward link is one composite signal transmitted to all mobiles at once; each mobile pulls out its own stream using its own code. In the frequency-time-code cube every channel is a layer along the *code* axis, using all of the frequency and all of the time. SST Ch 7



CDMA against FHMA in the same axes. CDMA users fill the whole frequency-time plane and are stacked along the code axis. FHMA users take one frequency each per time slot and swap around. SST Ch 7

## CDMA design considerations: self-jamming and the near-far problem 2 4 4+5

TOP 9/23 80 Ba 4, 72 Ash 2, 76 Bh 5+4, 71 Ma 3+4 and 2+2 · these two are the "limitations" half of the CDMA question

### Self-jamming 2

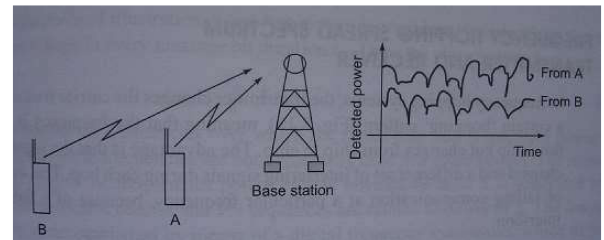
- Spreading sequences of different users are **approximately, not exactly, orthogonal**.
- So when the receiver de-spreads with one user's PN code, the other users' transmissions make **non-zero contributions** instead of cancelling to zero.
- Those residues appear as extra noise on the wanted signal.
- **The system jams itself**, which is why the noise floor rises with every added user and why capacity is soft rather than hard.
- Worse on the **reverse link**: mobiles are unsynchronised, so their codes cannot be held orthogonal. On the forward link all channels leave one base station in step, so Walsh codes stay orthogonal.
- Cure: longer codes (higher  $PG$ ), tight power control, and error control coding.

### Near-far problem 2

- Occurs when **an undesired user is detected at much higher power than the desired user**.
- Cause: mobile A sits beside the base station, mobile B is at the cell edge. Path loss differs by tens of dB.
- Since one user's transmission is another's noise, B's signal falls below the required SNR and **is not detectable at all**.
- It is **specific to CDMA**, because everyone shares one frequency at one time. FDMA, TDMA and FHMA separate users before power ever matters.

### (+) of CDMA, including over TDMA 4

- **Higher capacity** for the same spectrum, helped by voice activity gain and universal frequency reuse.
- **No frequency planning**, since  $N = 1$ .
- **Soft capacity**: extra users degrade quality gracefully instead of being blocked.
- **Soft handoff**, so no dropped calls at cell edges. TDMA can only do hard handoff.
- **Multipath is exploited**, not just tolerated, via the RAKE receiver.
- **Privacy** is built into the waveform.
- Lower average transmit power thanks to fast power control → longer battery life.



Near-far scenario. Mobile A is close in, mobile B is far out. The detected power from A swamps that from B at the base station receiver. SST Ch 7

### All possible mitigation techniques 4

1. **Power control** (the main answer). Every base station commands each mobile in its coverage area to adjust transmit power so that **all mobiles arrive at the base station at the same level**.
  - **Open loop**: mobile measures the forward link and sets its own power.
  - **Closed loop**: base station measures the reverse link and sends up/down commands, in IS-95 at **800 times per second**.
2. **Hybrid DS/FHMA**: adding frequency hopping means the strong and weak users are rarely on the same frequency at the same instant (§7.8).
3. **TCDMA**: only one user transmits per cell per slot, so there is no second user to swamp (§7.8).
4. **Multi-user detection / interference cancellation**: decode the strong user first, subtract it, then decode the weak one. verified/added
5. **Sectoring and smart antennas (SDMA)**, which separate the two users in space.

### (-) and limitations 4

- **Self-jamming**: codes are not perfectly orthogonal.
- **Near-far problem**, which forces very fast and accurate power control.
- Needs **tight, continuous power control** → complex, and a failure drops capacity sharply.
- Needs **network-wide timing**, in IS-95 taken from GPS.
- **Receiver is complex**: correlators, RAKE fingers, code acquisition and tracking.
- Capacity is **interference limited**, so it falls whenever the load in neighbouring cells rises.
- Wide bandwidth per carrier (1.25 MHz in IS-95), so a small operator cannot deploy it in a narrow allocation.

## 7.7 Implementation of CDMA: encoding and decoding

Explain the implementation of CDMA with its encoding and decoding parts, with a relevant example

7 8 2+6

**HOT 3/23** 82 Bb 7, 79 Ch 8, 77 Ch 2+6 · never answer this without numbers: the marks are in the worked chip arithmetic · two full worked versions in companion §5

The four rules the whole scheme rests on

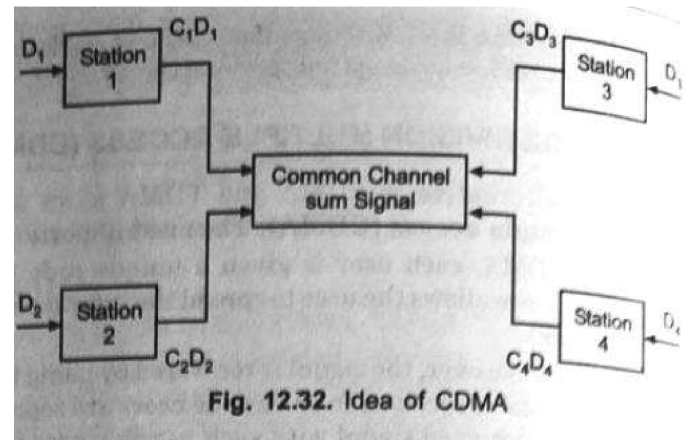
1. **Codes are orthogonal.** Multiply two different codes element by element, add up, and the result is 0.
2. **A code times itself** gives  $N$ , the number of elements (the number of stations).
3. **Data representation:** bit 1  $\rightarrow +1$ , bit 0  $\rightarrow -1$ , **silent**  $\rightarrow 0$ .
4. **Scaling:** multiplying a sequence by a number multiplies every element by that number.

**Encoding (transmit side)** 3

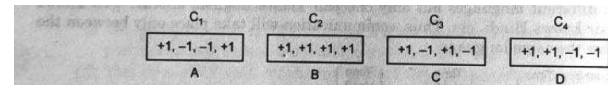
1. Assign each station a unique orthogonal chip sequence  $C_1 \dots C_N$ .
2. Each station computes  $d_i C_i$ , its data value times its own code.
3. All stations transmit at once. The channel **adds** the sequences element by element.
4. Result on the common channel is one composite sequence  $S = \sum_i d_i C_i$ .

**Decoding (receive side)** 3

1. Every station receives the same composite  $S$ .
2. To read station  $k$ , multiply  $S$  by  $C_k$  element by element.
3. **Add all the products** to get one number.
4. **Divide by  $N$ .** Every other user's term vanished by orthogonality, so what is left is  $d_k$ .
5.  $+1 \rightarrow$  bit 1,  $-1 \rightarrow$  bit 0,  $0 \rightarrow$  that station was silent.

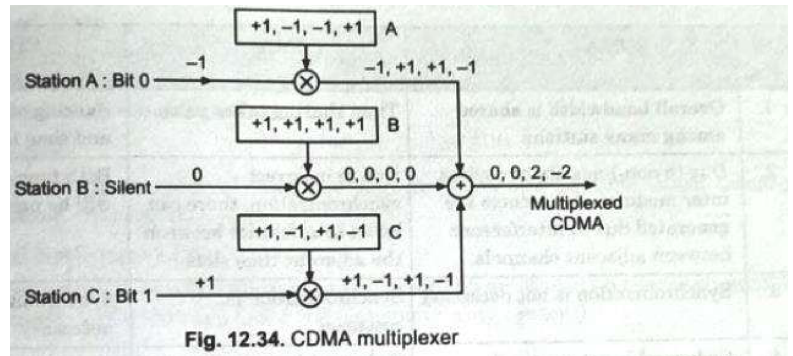


The idea of CDMA. Four stations each put  $C_i D_i$  onto one common channel, which carries the sum. SST Ch 7, Fig 12.32

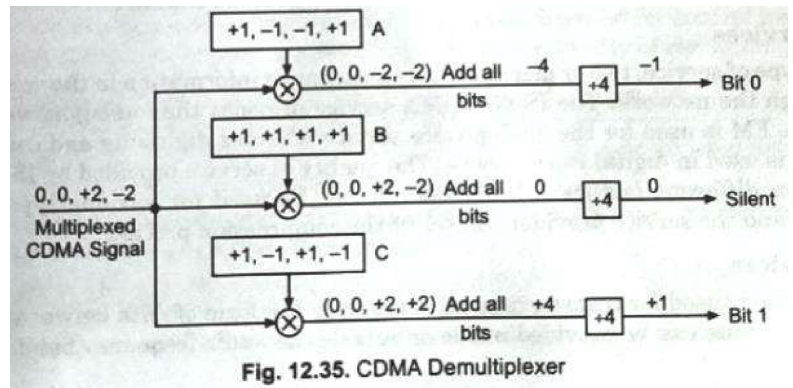


### The worked example to reproduce 4

- Codes:  $C_A = (+1, -1, -1, +1)$ ,  
 $C_B = (+1, +1, +1, +1)$ ,  $C_C = (+1, -1, +1, -1)$ .
- A sends 0  $\rightarrow -1$ , B is silent  $\rightarrow 0$ , C sends 1  $\rightarrow +1$ .
- Encode:  $-1 \cdot C_A = (-1, +1, +1, -1)$ ,  
 $0 \cdot C_B = (0, 0, 0, 0)$ ,  $+1 \cdot C_C = (+1, -1, +1, -1)$ .
- Sum on the channel:  $S = (0, 0, +2, -2)$ .
- Decode A:  $S \times C_A = (0, 0, -2, -2)$ , sum =  $-4$ ,  
 $-4/4 = -1 \rightarrow$  **bit 0** ✓
- Decode B:  $S \times C_B = (0, 0, +2, -2)$ , sum =  $0$ ,  
 $0/4 = 0 \rightarrow$  **silent** ✓
- Decode C:  $S \times C_C = (0, 0, +2, +2)$ , sum =  $+4$ ,  
 $+4/4 = +1 \rightarrow$  **bit 1** ✓
- Always end by checking all three decode correctly.** That is the proof the codes are orthogonal, and it is where the last mark sits.



CDMA multiplexer (encoder). A sends bit 0 ( $-1$ ), B is silent ( $0$ ), C sends bit 1 ( $+1$ ). The adder puts  $(0, 0, +2, -2)$  on the channel. SST Ch 7, Fig 12.34



CDMA demultiplexer (decoder). The one multiplexed sequence  $(0, 0, +2, -2)$  enters three correlator branches. Each multiplies by its own code, adds all the bits, and divides by 4 to recover  $-1, 0$  and  $+1$ . SST Ch 7, Fig 12.35

### Steps to satisfy the condition for a Hadamard code; construct the 8 by 8 Hadamard code 3 4 3+5

**HOT 3/23** 82 Bh 2+3, 80 Bh 4, 79 Ch 3+5 · full construction with every intermediate matrix in companion §4

#### Conditions a Hadamard matrix must satisfy 3

- Square, of order  $N$ , with every entry  $+1$  or  $-1$ .
- Any two distinct rows are orthogonal:** their element-wise product sums to 0.
- Equivalently  $H_N H_N^T = N I_N$ .
- Order must be  $N = 1, 2$ , or a **multiple of 4**.
- Rows are the **codewords**. Length  $N$ , so  $N$  codewords carry  $k = \log_2 N$  information bits.
- Minimum distance  $d_{\min} = N/2$ , so it corrects up to  $N/4 - 1$  errors.

#### Construction, the three steps to write down 5

- Start from  $H_1 = [1]$ , or from  $H_2 = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ .
- Apply the **Sylvester recursion**, doubling the order each time:

$$H_{2N} = \begin{bmatrix} H_N & H_N \\ H_N & -H_N \end{bmatrix}$$

- Repeat until the wanted order.  $H_2 \rightarrow H_4 \rightarrow H_8$ .
- Verify: pick any two rows, multiply, add. **Must give 0**.
- For CDMA use, map  $+1 \rightarrow 0$  and  $-1 \rightarrow 1$  to get the binary **Walsh codes**.

**Where it is used:** the rows of  $H_{64}$  are the 64 Walsh functions that separate the forward channels of IS-95 CDMA. See also Ch 6, block codes.

## 7.8 Hybrid spread spectrum multiple access techniques

Explain any two hybrid spread spectrum multiple access techniques, with (+) and (-); which of them mitigate the near-far problem 4 6 7

**TOP 9/23** 81 Bh 6, 80 Bh 4, 80 Ch 6, 76 Bh 4, 75 Bh 6, 74 Bh 6, 72 Ash 6, 71 Bh 7, 71 Ma 2+2 · 9 of 23 papers, and it is nearly always bolted onto the FHMA question · the question usually says "any two", so learn DS/FHMA and TCDMA and the near-far answer comes free

### 1. Hybrid FDMA/CDMA (FCDMA)

- An alternative to plain DS-CDMA.
- Available wideband spectrum is cut into a number of sub-spectra of smaller bandwidth.
- Each sub-channel becomes its own narrowband CDMA system.
- Different users get different sub-spectrum bandwidths to suit their needs.
- ✓ Lets an operator deploy CDMA in a narrow or fragmented allocation, instead of needing one contiguous wide band.
- ✓ Bandwidth on demand: a data user can be given a wider sub-spectrum.
- × Each sub-system has a lower processing gain than the full-band system, so interference rejection is worse.
- × Guard bands between sub-spectra waste spectrum, and frequency planning returns.
- × Does not solve near-far, since inside a sub-channel it is still CDMA.

### 2. Hybrid direct sequence / frequency hopped (DS/FHMA)

- A direct sequence modulated signal whose centre frequency is made to hop periodically in a pseudorandom fashion.
- So each burst is spread twice: by the PN chip code, and again by being placed on a pseudorandom carrier.
- ✓ Avoids the near-far problem, because frequency diversity is introduced: a strong nearby user and a weak far user are rarely on the same hop.
- ✓ Doubles the resistance to jamming and to frequency-selective fading.
- ✓ Keeps the multipath resolution of DS while gaining FH's collision avoidance.
- × Not adaptable to soft handoff: the FH base station receiver would have to synchronise to multiple hopped signals at once.
- × Most complex of the four to build and to synchronise.

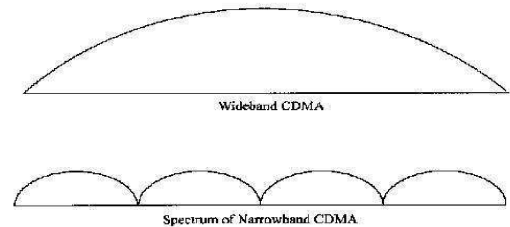


Figure 9.6  
Spectrum of wideband CDMA compared to the spectrum of a hybrid, frequency division, direct sequence multiple access.

FCDMA. One wideband CDMA spectrum (top) against the same allocation split into several narrowband CDMA sub-spectra (bottom). SST Ch 7, Fig 9.6

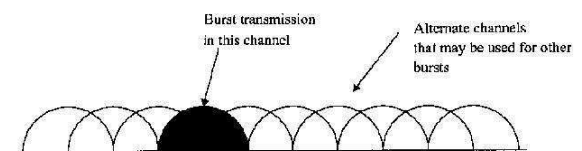


Figure 9.7  
Frequency spectrum of a hybrid FH/DS system.

Hybrid FH/DS spectrum. Each arch is one direct-sequence burst; the filled arch is the channel in use for this hop and the rest are available to other bursts. SST Ch 7, Fig 9.7

### 3. Time division CDMA (TCDMA)

- Different cells are allocated **different spreading codes**.
- Within a cell, **only one user per cell is allotted a particular time slot**.
- So at any instant **only one user is transmitting in each cell**.
- On handoff, the user's spreading code is **changed to that of the new cell**.
- ✓ **Avoids the near-far effect**, because there is no second user in the cell to swamp the first.
- ✓ No self-jamming inside a cell, since only one code is active.
- ✓ Power control requirements are far looser.
- × Throws away CDMA's soft capacity: capacity is now hard limited by slots.
- × Needs TDMA-grade timing and a code switch at every handoff.

### 4. Time division frequency hopping (TDFH)

- The subscriber **hops to a new frequency at the start of every new TDMA frame**, following a pseudorandom hopping sequence.
- At each time slot the mobile is hopped to a new frequency.
- **Adopted in GSM**, where the hopping sequence is predefined and the subscriber may hop only on certain permitted frequencies.
- ✓ Avoids **severe deep frequency-selective fades**: a bad frequency is abandoned at the next frame.
- ✓ Avoids **severe co-channel interference**, which is otherwise fixed for the whole call.
- ✓ Averages interference across all users, so no single user is persistently unlucky.
- × Adds a synthesizer that must retune within the guard time.
- × Hopping sequences must be planned across cells to stop repeated collisions.

If the question says "which mitigate the near-far problem", the answer is **DS/FHMA and TCDMA** (75 Bh, 74 Bh, 71 Ma). FCDMA and TDFH do not: FCDMA is still CDMA inside each sub-band, and TDFH is a TDMA system that never had the problem. Say so explicitly, it is worth a mark.

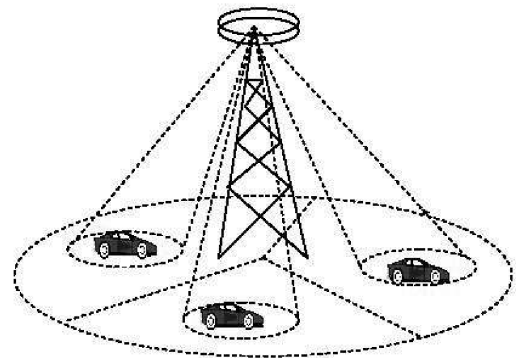
## 7.9 SDMA: space division multiple access

### What is space division multiple access? 2 6

PIN 2/23 75 Bh 2, 71 Bh 2+6 · always a 2-mark opener to a 6-mark hybrid-SSMA or compare-the-schemes rider

#### Principle 2

- **Controls the radiated energy for each user in space**.
- Serves different users with **spot beam antennas**, each beam aimed at one user or one small area.
- Areas covered by different beams may reuse **the same frequency** (in a TDMA or CDMA system) or use **different frequencies** (in an FDMA system).
- **Sectorized antennas are the primitive application of SDMA**, splitting a cell into 120° or 60° sectors (Ch 2).
- Adaptive / smart antennas are the full version: the beam **tracks the mobile** and steers a null toward interferers.
- Modern form is **MIMO**: many antennas at both ends create parallel spatial channels in the same band. Adding antennas adds channels.



**SDMA.** One base station forms several spot beams, each serving a different user in a different direction on the same channel. SST Ch 7

(+)

- **Frequency reuse *inside a single cell***, which no other scheme offers.
- Multiplies capacity by roughly the number of beams.
- Cuts co-channel interference: energy is not sprayed where no user is.
- Range extends, because antenna gain rises with beam narrowness.
- Reduces the near-far problem, by separating the two users spatially.
- **Layers on top of anything:** SDMA is used *with* FDMA, TDMA or CDMA, never instead of them.

(-)

- Needs an **antenna array** and real-time beamforming hardware at the base station.
- Beams must **track moving users**, so direction of arrival must be estimated continuously.
- Two users in the **same direction** cannot be separated by space alone.
- Handoff between beams adds a second layer of handoff to manage.
- Expensive, so it is deployed in high-density cells first.

## 7.10 Comparing the schemes

Compare FDMA, TDMA and CDMA: state the (+) and (-) of each 2 4 4+4

**TOP 9/23** 73 Ma 4+4, 81 Ba 4, 78 Ch 4, 72 Ma 2, 70 Ma 4, 71 Bh 6, 76 Bh 5, 71 Ma 4, 70 Bh 6 · write the table, then two bullets per scheme. A table alone is worth most of the marks

|                     | FDMA                              | TDMA                                | CDMA                       |
|---------------------|-----------------------------------|-------------------------------------|----------------------------|
| Users separated by  | frequency                         | time                                | code                       |
| Bandwidth per user  | narrow (25–30 kHz)                | full carrier, part of the time      | full band, all of the time |
| Signal type         | narrowband                        | narrowband                          | wideband                   |
| Transmission        | continuous                        | bursty                              | continuous                 |
| Duplexer in handset | <b>required</b>                   | not required                        | required (FDD)             |
| Equalizer           | rarely needed                     | <b>always needed</b>                | RAKE instead               |
| Overhead bits       | lowest                            | <b>highest</b> (sync, guard, trail) | moderate                   |
| Synchronisation     | none needed                       | tight, network wide                 | tight, plus power control  |
| Capacity limit      | <b>hard</b> (channels)            | <b>hard</b> (slots)                 | <b>soft</b> (noise floor)  |
| Frequency reuse     | 7 typically                       | 4 to 7                              | <b>1</b>                   |
| Handoff             | hard                              | hard, MAHO assisted                 | <b>soft</b>                |
| Main weakness       | idle channels wasted, IM products | overhead and equalizer cost         | near-far, self-jamming     |
| Example system      | AMPS                              | GSM, IS-136                         | IS-95, WCDMA               |

### FDMA in two lines

- ✓ Simple, no sync, no equalizer, low overhead.
- × Idle channels wasted, needs duplexer, tight filters, IM at the base station.

### TDMA in two lines

- ✓ No duplexer, low battery drain, MAHO, bandwidth on demand, cheaper cell site.
- × Equalizer compulsory, high overhead, needs precise network timing.

### CDMA in two lines

- ✓ Soft capacity, soft handoff,  $N = 1$ , multipath used as diversity, private.
- × Near-far and self-jamming, so it lives or dies on power control.

### The one-sentence verdict

- **FDMA wastes spectrum, TDMA wastes bits, CDMA wastes power.** Each scheme pays for multiple access in a different currency.

## 7.11 Standards for wireless local area networks no lecture note covers this

### Write short notes on Wireless Local Area Network (WLAN) and WiFi 3 5

**PIN 2/23** 75 Bh 3 (WLAN), 76 Bh 5 (WiFi) · verified/added a syllabus topic with zero coverage in any of the three lecture decks · WiMAX and LTE belong to Ch 8, do not answer them here

#### What a WLAN is 3

- A local area network in which the links between stations are **radio**, not cable.
- Covers a building or campus, tens to hundreds of metres, not a cellular service area.
- Runs in **unlicensed ISM / U-NII bands**: 2.4 GHz, 5 GHz and now 6 GHz.
- Standardised as the **IEEE 802.11** family. **Wi-Fi is the trade name** of the Wi-Fi Alliance for certified 802.11 equipment.
- Two topologies:
  - **Infrastructure (BSS)**: stations talk through an **access point (AP)**, which bridges to the wired LAN.
  - **Ad hoc (IBSS)**: stations talk peer to peer, with no AP.
- Several BSSs joined by a distribution system form an **ESS**, which is what gives roaming across a building.

#### How the medium is shared 2

- Multiple access is **contention based**, not assigned. **This is the key difference from FDMA / TDMA / CDMA.**
- **CSMA/CA**: listen first, and if the channel is busy back off a random time.
- **Collision avoidance, not detection**: a radio cannot hear while transmitting, so 802.11 cannot use CSMA/CD like Ethernet.
- Every correctly received frame is **acknowledged**. No ACK is taken as a collision.
- Optional **RTS/CTS** handshake solves the **hidden node problem**, where two stations can each hear the AP but not each other.
- **DCF** is the contention mode, always present. **PCF** is an optional polled, contention-free mode.
- Security: WEP (broken), then WPA, WPA2 (AES-CCMP), WPA3.

#### The 802.11 standards worth naming 2 verified/added

| Standard  | Year | Band            | Peak rate | What it added                                    |
|-----------|------|-----------------|-----------|--------------------------------------------------|
| 802.11    | 1997 | 2.4 GHz         | 1–2 Mbps  | the original, FHSS or DSSS                       |
| 802.11b   | 1999 | 2.4 GHz         | 11 Mbps   | DSSS with CCK, the first popular one             |
| 802.11a   | 1999 | 5 GHz           | 54 Mbps   | <b>OFDM</b> , 52 sub-carriers, clean band        |
| 802.11g   | 2003 | 2.4 GHz         | 54 Mbps   | 802.11a's OFDM in the 2.4 GHz band               |
| 802.11n   | 2009 | 2.4 / 5 GHz     | 600 Mbps  | <b>MIMO</b> , 40 MHz channels, frame aggregation |
| (Wi-Fi 4) |      |                 |           |                                                  |
| 802.11ac  | 2013 | 5 GHz           | ~6.9 Gbps | <b>MU-MIMO</b> , 160 MHz channels, 256-QAM       |
| (Wi-Fi 5) |      |                 |           |                                                  |
| 802.11ax  | 2019 | 2.4 / 5 / 6 GHz | ~9.6 Gbps | <b>OFDMA</b> , better in dense crowds, 1024-QAM  |
| (Wi-Fi 6) |      |                 |           |                                                  |

Related standards in the same family, worth one line each: **802.15.1** Bluetooth (WPAN, FHMA over 79 channels), **802.15.4** Zigbee, **802.16** WiMAX (WMAN, answered in Ch 8).

#### WLAN (+)

- **Mobility** inside the coverage area.
- **No cabling**, so installation is fast and cheap, and works in old buildings.
- **Scalable**: add a station without touching the infrastructure.
- **Unlicensed spectrum**, so no spectrum fee.

#### WLAN (-)

- **Shared, contended** medium → throughput falls as users are added.
- **Unlicensed** band → interference from microwave ovens, Bluetooth, neighbouring networks, with no legal recourse.
- **Short range**, and walls attenuate heavily at 5 GHz and above.
- **Security** is harder than on a cable, since the signal leaves the building.

## 7.12 Last-minute recall for this chapter

### Must memorise

- FDMA channels:  $N = \frac{B_t - 2B_{\text{guard}}}{B_c}$ . **The 2 is not optional.**
- TDMA channels:  $N = \frac{m(B_{\text{tot}} - 2B_{\text{guard}})}{B_c}$ .
- Frame efficiency:  $\eta_f = \left(1 - \frac{b_{OH}}{b_T}\right) \times 100\%$ , **GSM = 74.24 %**.
- GSM slot =  $6 + 8.25 + 26 + 2(58) = 156.25$  bits; frame =  $8 \times 156.25 = 1250$  bits.
- Processing gain  $PG = B_{ss}/B = T_b/T_c = N$ .
- Hadamard:  $H_{2N} = \begin{bmatrix} H_N & H_N \\ H_N & -H_N \end{bmatrix}$ , order 1, 2 or a multiple of 4,  $d_{\min} = N/2$ .
- CDMA decode: multiply by the code, add all, **divide by  $N$** .
- Data mapping:  $1 \rightarrow +1$ ,  $0 \rightarrow -1$ , silent  $\rightarrow 0$ .
- Fast hopping = hop rate  $\geq$  symbol rate. Slow hopping = hop rate  $\leq$  symbol rate.
- Pairings: AMPS FDMA/FDD, GSM TDMA/FDD, DECT FDMA/TDD, IS-95 CDMA/FDD.
- Near-far cures: **power control, DS/FHMA, TCDMA**, multi-user detection, SDMA.
- IS-95 closed-loop power control runs at **800 Hz**.

### Most repeated, in order

1. FHMA principle + transmitter and receiver +Fig **TOP 10/23**
2. Two hybrid SSMA techniques, and which fix near-far **TOP 9/23**
3. CDMA features, (+) and limitations **TOP 9/23**
4. Define multiple access + compare FDMA/TDMA/CDMA **TOP 9/23**
5. FDMA channel-count numerical **HOT 4/23**
6. CDMA encoding and decoding +Eg **HOT 3/23**
7. Construct the  $8 \times 8$  Hadamard code **HOT 3/23**
8. SSMA variants, DS against FH **HOT 3/23**
9. SDMA **PIN 2/23** · WLAN and WiFi **PIN 2/23**
10. TDMA numericals: N-TDMA users, GSM frame efficiency **PIN 2/23**

**The single highest-value hour** you can spend on this subject is drawing the FHMA transmitter and receiver until they come out from memory. Ten of the twenty-two papers pay for it, usually at 4 to 7 marks, and the hybrid-SSMA question that follows it is worth another 4 to 6.

**Five numericals live here**, all worked in the companion PDF: FDMA channel count (3 variants), N-TDMA users per cluster, GSM frame efficiency, the  $H_8$  construction, and CDMA encode/decode.

### Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

| Short    | Means                                   | Short    | Means                                      |
|----------|-----------------------------------------|----------|--------------------------------------------|
| BS       | base station                            | BTS      | base transceiver station                   |
| MS       | mobile station                          | MSC      | mobile switching centre                    |
| BSC      | base station controller                 | MSK      | minimum shift keying                       |
| CCI      | co-channel interference                 | ACI      | adjacent channel interference              |
| SIR, S/I | signal to interference ratio            | SNR, S/N | signal to noise ratio                      |
| GoS      | grade of service                        | MAHO     | mobile assisted handoff                    |
| FCA      | fixed channel assignment                | DCA      | dynamic channel assignment                 |
| LOS      | line of sight                           | NLOS     | non line of sight                          |
| FDMA     | frequency division multiple access      | TDMA     | time division multiple access              |
| CDMA     | code division multiple access           | OFDM     | orthogonal frequency division multiplexing |
| FDD      | frequency division duplex               | TDD      | time division duplex                       |
| GSM      | global system for mobile communications | AMPS     | advanced mobile phone system               |
| PCS      | personal communication system           | DAS      | distributed antenna system                 |
| BCCH     | broadcast control channel               | EIRP     | effective isotropic radiated power         |
| ERP      | effective radiated power                | RF       | radio frequency                            |
| Tx, TX   | transmitter                             | Rx, RX   | receiver                                   |

| Short      | Means                                       | Short         | Means                                      |
|------------|---------------------------------------------|---------------|--------------------------------------------|
| ISI        | intersymbol interference                    | PDP           | power delay profile                        |
| DFE        | decision feedback equalizer                 | MLSE          | maximum likelihood sequence estimation     |
| LMS        | least mean square                           | RLS           | recursive least squares                    |
| MSE        | mean square error                           | MMSE          | minimum mean square error                  |
| ZF         | zero forcing                                | MRC           | maximal ratio combining                    |
| EGC        | equal gain combining                        | BER           | bit error rate                             |
| PCM        | pulse code modulation                       | DPCM          | differential PCM                           |
| ADPCM      | adaptive differential PCM                   | LPC           | linear predictive coder                    |
| RPE-LTP    | regular pulse excited, long term prediction | CELP          | code excited linear prediction             |
| REL P      | residual excited linear prediction          | MPE           | multipulse excited                         |
| VSELP      | vector sum excited linear prediction        | ATC           | adaptive transform coding                  |
| SBC        | sub-band coding                             | DCT           | discrete cosine transform                  |
| FEC        | forward error correction                    | ARQ           | automatic repeat request                   |
| BCH        | Bose–Chaudhuri–Hocquenghem code             | RS            | Reed–Solomon code                          |
| TCM        | trellis coded modulation                    | PN            | pseudo-noise                               |
| MIMO       | multiple input multiple output              | OFDMA         | orthogonal FDM access                      |
| Short      | Means                                       | Short         | Means                                      |
| QAM        | quadrature amplitude modulation             | PSK           | phase shift keying                         |
| GMSK       | Gaussian minimum shift keying               | IMT           | international mobile telecommunications    |
| ASK        | amplitude shift keying                      | FSK           | frequency shift keying                     |
| BPSK       | binary phase shift keying                   | QPSK          | quadrature phase shift keying              |
| OQPSK      | offset (staggered) QPSK                     | DPSK,         | differential (binary) PSK                  |
|            |                                             | DBPSK         |                                            |
| CPFSK      | continuous phase FSK                        | MFSK,         | M-ary FSK, M-ary PSK                       |
|            |                                             | MPSK          |                                            |
| DSSS       | direct sequence spread spectrum             | FHSS          | frequency hopped spread spectrum           |
| SS, SSM    | spread spectrum (modulation)                | PG            | processing gain                            |
| PAPR       | peak-to-average power ratio                 | CP            | cyclic prefix                              |
| FFT, IFFT  | (inverse) fast Fourier transform            | ICI           | inter-carrier interference                 |
| PSD        | power spectral density                      | CFO, SFO      | carrier, sampling frequency offset         |
| NRZ, RZ    | non-return, return to zero                  | AMI           | alternate mark inversion                   |
| AM, FM, PM | amplitude, frequency, phase modulation      | DSB, SSB, VSB | double, single, vestigial sideband         |
| S/P, P/S   | serial-parallel converter                   | D/A, A/D      | digital-analog converter                   |
| LPF, BPF   | low pass, band pass filter                  | ISM           | industrial, scientific, medical band       |
| SC-FDMA    | single carrier FDMA                         | <i>BT</i>     | GMSK filter bandwidth $\times$ symbol time |

| Short     | Means                                                  | Short    | Means                                               |
|-----------|--------------------------------------------------------|----------|-----------------------------------------------------|
| PSTN      | public switched telephone network                      | VoIP     | voice over IP                                       |
| FVC, RVC  | forward, reverse voice channel                         | FCC, RCC | forward, reverse control channel                    |
| SSMA      | spread spectrum multiple access                        | SDMA     | space division multiple access                      |
| FHMA      | frequency hopped multiple access                       | DSMA     | direct sequence multiple access (= CDMA)            |
| FCDMA     | hybrid FDMA/CDMA                                       | DS/FHMA  | hybrid direct sequence / frequency hopped           |
| TCDMA     | time division CDMA                                     | TDFH     | time division frequency hopping                     |
| IM        | intermodulation (product)                              | MU-MIMO  | multi-user MIMO                                     |
| WLAN      | wireless local area network                            | WMAN     | wireless metropolitan area network                  |
| WPAN      | wireless personal area network                         | AP       | access point                                        |
| BSS, IBSS | basic service set, independent (ad hoc) BSS            | ESS      | extended service set                                |
| CSMA/CA   | carrier sense multiple access with collision avoidance | CSMA/CD  | ... with collision detection                        |
| DCF       | distributed coordination function                      | PCF      | point coordination function                         |
| RTS/CTS   | request to send / clear to send                        | CCK      | complementary code keying                           |
| WEP, WPA  | wired equivalent privacy, WiFi protected access        | U-NII    | unlicensed national information infrastructure band |
| IoT       | internet of things                                     | mmWave   | millimetre wave                                     |
| eMBB      | enhanced mobile broadband                              | URLLC    | ultra-reliable low latency comm.                    |

| Short   | Means                          | Short  | Means                                    |
|---------|--------------------------------|--------|------------------------------------------|
| mMTC    | massive machine type comm.     | NOMA   | non-orthogonal multiple access           |
| D2D     | device to device               | C-RAN  | cloud radio access network               |
| PCN     | personal communication network | SMS    | short message service                    |
| attr(s) | attribute(s)                   | no.    | number                                   |
| freq    | frequency                      | info   | information                              |
| ops     | operations                     | config | configuration                            |
| rm      | remove                         | vs, Vs | versus                                   |
| w/, w/o | with, without                  | Eg     | example / for example                    |
| +Fig    | draw a figure                  | (I)    | importance / need / significance         |
| →       | leads to / therefore           | +      | and                                      |
| BSS     | base station subsystem         | NSS    | network and switching subsystem          |
| OSS     | operation support subsystem    | OMC    | operation and maintenance centre         |
| HLR     | home location register         | VLR    | visitor location register                |
| AuC     | authentication centre          | EIR    | equipment identity register              |
| CEIR    | central EIR                    | ME     | mobile equipment                         |
| SIM     | subscriber identity module     | IMSI   | international mobile subscriber identity |

| Short       | Means                                     | Short          | Means                                  |
|-------------|-------------------------------------------|----------------|----------------------------------------|
| IMEI        | international mobile equipment identity   | TMSI           | temporary mobile subscriber identity   |
| Um, Abis, A | the three GSM interfaces                  | SS7            | signalling system no. 7                |
| ARFCN       | absolute radio frequency channel number   | TN, TS         | time slot number, time slot            |
| TCH         | traffic channel                           | TCH/FS, TCH/HS | full rate, half rate speech TCH        |
| CCH         | control channel                           | CCCH           | common control channel                 |
| DCCH        | dedicated control channel                 | SDCCH          | stand-alone dedicated control channel  |
| SACCH       | slow associated control channel           | FACCH          | fast associated control channel        |
| FCCH        | frequency correction channel              | SCH            | synchronisation channel                |
| PCH         | paging channel                            | RACH           | random access channel                  |
| AGCH        | access grant channel                      | TB, GP, SF     | tail bits, guard period, stealing flag |
| VAD         | voice activity detector                   | DTX            | discontinuous transmission             |
| DCS 1800    | digital cellular system, the 1800 MHz GSM | CRC            | cyclic redundancy check                |
| ESN         | electronic serial number                  | PCB            | power control bit                      |
| Mcps, Ksps  | mega-chips, kilo-symbols per second       | PRAT           | paging channel rate field              |
| EVRC        | enhanced variable rate coder              | PDSN           | packet data serving node               |
| PCF (CDMA)  | packet control function                   | RNC            | radio network controller               |

| Short      | Means                                   | Short            | Means                                |
|------------|-----------------------------------------|------------------|--------------------------------------|
| EPS, EPC   | evolved packet system, core             | eNodeB           | LTE base station                     |
| MME        | mobility management entity              | S-GW, P-GW       | serving, packet data network gateway |
| HSS        | home subscriber server                  | PCRF             | policy and charging rules function   |
| IMS, VoLTE | IP multimedia subsystem, voice over LTE | CoMP             | coordinated multipoint               |
| ITU-R      | ITU radiocommunication sector           | WRC              | world radiocommunication conference  |
| NTA        | Nepal Telecommunications Authority      | UGS, rtPS, nrtPS | WiMAX QoS service classes            |

**Standard and system names** (proper names, kept as printed): **AMPS** advanced mobile phone system · **ETACS** extended total access communication system · **NMT** Nordic mobile telephone · **NTT** Nippon Telegraph and Telephone · **IMTS** improved mobile telephone service · **PDC** Pacific digital cellular · **IS-95** cdmaOne · **IS-136** digital AMPS · **HSCSD** high speed circuit switched data · **GPRS** general packet radio service · **EDGE** enhanced data rates for GSM evolution · **W-CDMA** wideband CDMA · **UMTS** universal mobile telecommunications system · **cdma2000 1xRTT / 1xEV-DO / 1xEV-DV** the cdma2000 3G family · **LTE, LTE-A** long term evolution (advanced) · **WiMAX** IEEE 802.16 · **WiFi** IEEE 802.11 · **DECT** digital enhanced cordless telecommunications · **CT2** cordless telephone 2 · **PHS** personal handyphone system · **USDC** US digital cellular (IS-136) · **JDC** Japanese digital cellular (later PDC) · **Bluetooth** IEEE 802.15.1 · **Zigbee** IEEE 802.15.4 · **PACS** personal access communications system · **POCSAG, FLEX** paging protocols · **CDPD** cellular digital packet data · **Mobitex** a dedicated packet data system · **TETRA** terrestrial trunked radio · **DAB** digital audio broadcasting · **DVB-T** digital video broadcasting, terrestrial · **ADSL** asymmetric digital subscriber line · **5G NR** 5G new radio.

**Three clashes to watch:** **FCC** is the *forward control channel* throughout these notes; the US regulator is always written out as the Federal Communications Commission, and **FCCH** is GSM's frequency correction channel, a different thing again. **BCH** is the Bose–Chaudhuri–Hocquenghem block code in **Ch 6** and GSM's *broadcast channel* in **Ch 8**. **SCH** is GSM's synchronisation channel, never a sub-channel.

**Exam-paper month codes** (used in the year tags): Ba = Baishakh · Bh = Bhadra · Ash = Ashwin · Ma = Magh · Ch = Chaitra.

**Bold** year tags are Regular exams, plain are Back exams. Monospace year tags are BEI (EX 715); the rest are BEX (EX 751).

**Tier chips** over 23 papers, 2070–2082 BS: **TOP /23** = topic asked in 6 or more papers · **HOT /23** = 3 to 5 papers · **PIN /23** = 1 or 2 papers but worth  $\geq 6$  marks. **Mark chips** such as **2+5** are the marks that question actually carried. **syllabus, no PYQ yet** marks a syllabus topic that has never been examined but is written up anyway. **verified/added** marks anything researched and added that was not in the source notes.